

Content Management in Health Care:

AI-Driven Policy Interpretation for Billing and Coding Compliance

Hariprakash Karthikeyan | University of Rochester | June 2026

Concept Definition

Healthcare content management is the systematic organization and retrieval of regulatory policy documents that govern medical payment and operations. In Medicare, this includes Local Coverage Determinations (LCDs), National Coverage Determinations (NCDs), payer-provider contracts, and standard documentation requirements published by the Centers for Medicare and Medicaid Services (CMS). These documents define which services are covered, how they must be billed, and what paperwork must exist for a claim to be paid. Unlike general content management, misapplication here carries direct financial and legal consequences.

Understanding the commercial context requires mapping the payment chain. CMS publishes policy documents. Durable medical equipment (DME) suppliers must read and apply those documents correctly to submit claims to Medicare. Cotiviti operates as a payment integrity contractor between supplier claims and Medicare payment: its systems identify incorrect payments before or after they occur. In FY2025, Medicare fee-for-service improper payments totaled \$28.8 billion at a 6.55% error rate. DME alone carried a 24.1% improper payment rate, totaling \$2.3 billion. Critically, 64% of traditional Medicare improper payments are attributable to documentation gaps and policy misapplication, not fraud (CMS, 2026). These errors originate at the supplier level. A tool that helps suppliers apply policy correctly at the source directly reduces the errors that Cotiviti is contracted to address.

Relevant Trends

Retrieval-Augmented Generation

Large language models paired with document retrieval pipelines enable citation-grounded answers from private corpora without retraining. Lewis et al. (2020) demonstrated that RAG architectures outperform parametric-only models on knowledge-intensive tasks. Applied to healthcare policy, the model retrieves the relevant LCD section and answers from it, rather than relying on training data that may reflect outdated rules.

Hybrid Lexical-Semantic Retrieval

Production retrieval systems combine keyword search (BM25) with vector search to maximize recall over technical documents. Medicare billing policy contains exact codes such as E2102 and the KF modifier that semantic embeddings alone can miss when user phrasing differs from source text. Reciprocal Rank Fusion merges both ranked lists, and cross-encoder reranking re-scores the top candidates with full query context (Nogueira and Cho, 2019).

AI Adoption in Payer Operations

As of 2026, 94% of payers report AI adoption in some form, with full-scale adoption rising from 8% in 2024 to 34% in 2025 (HealthEdge, 2026; Prosper AI, 2026). This is running alongside tightening regulation: over 250 healthcare AI bills have been introduced across 34 states, several requiring disclosure of AI involvement in claims decisions (Nixon Law Group, 2026). Tools in this space must be auditable and source-attributed, not merely fast.

Opportunities and Threats

Opportunities

- **New revenue stream:** Cotiviti currently earns revenue catching errors after claims are submitted. A policy Q&A tool creates a parallel model: selling a subscription-based compliance tool to DME suppliers before claims are submitted. Thousands of suppliers face this problem with no purpose-built solution available today.
- **Stronger core product:** When suppliers submit correct claims from the start, Cotiviti's audit systems can focus on true anomalies rather than administrative errors, improving the accuracy and value of existing payment integrity products.

- **Policy change monitoring:** The same retrieval architecture can compare document versions, detect what changed in a CMS revision, and deliver proactive alerts to clients before new rules take effect, a defensible high-value service in a market where regulatory velocity is increasing.
- **Contract intelligence:** Payer-provider contracts contain fee schedules, carve-outs, and policy references. The same platform can index these documents to resolve payment disputes and surface underpayments, extending reach beyond Medicare.

Threats

- **Hallucination and compliance risk:** Even citation-grounded systems can produce imprecise outputs when retrieved content is insufficient. In billing, a fabricated code is a compliance violation. Confidence scoring and human review protocols are required before deployment.
- **Data staleness:** LCDs are revised regularly. An index without an automated re-ingestion pipeline tied to CMS revision dates will silently serve outdated guidance and become a liability.
- **Regulatory exposure:** Automated policy interpretation may be construed as legal advice under new state AI-disclosure laws. Clear disclaimers, audit logging, and governance protocols are necessary before production use.
- **Licensing constraints:** CPT and HCPCS code descriptors are AMA-licensed, limiting what can be freely ingested or redistributed in a retrieval corpus without a separate agreement.

Strategic Options for Cotiviti

Three strategic actions are proposed in priority order.

1. **Build and sell a managed RAG platform for policy documents.** Develop a scalable pipeline that continuously ingests CMS LCD and NCD updates, indexes them with hybrid retrieval, and exposes a citation-grounded Q&A interface to DME suppliers as a SaaS subscription. This creates a new recurring revenue stream while reducing the error volume that reaches Cotiviti's existing audit systems. The DME category, at a 24.1% improper payment rate, is the logical starting point.
2. **Launch a policy change monitoring service.** Use document comparison and LLM summarization to detect changes between LCD versions and deliver proactive alerts to clients before effective dates. Compliance teams currently discover rule changes through denied claims. First-mover advantage here is material and not easily replicated.
3. **Establish an AI governance framework first.** Define confidence thresholds, human review escalation paths, source attribution standards, and audit logging before broader deployment. Cotiviti's credibility in payment integrity depends on AI recommendations being explainable and traceable, not only accurate in testing.

References

1. Centers for Medicare and Medicaid Services. (2026, January). Medicare improper payments fact sheet. CMS.gov. <https://www.cms.gov/newsroom/fact-sheets/medicare-improper-payments>
2. Centers for Medicare and Medicaid Services. (2025). Glucose monitors (Local Coverage Determination L33822). Medicare Coverage Database. <https://www.cms.gov/medicare-coverage-database/view/lcd.aspx?LCDId=33822>
3. Centers for Medicare and Medicaid Services. (2025). Glucose monitor: Policy article (A52464). Medicare Coverage Database. <https://www.cms.gov/medicare-coverage-database/view/article.aspx?articleId=52464>
4. Centers for Medicare and Medicaid Services. (2024). Standard documentation requirements for all claims submitted to DME MACs (A55426). Medicare Coverage Database. <https://www.cms.gov/medicare-coverage-database/view/article.aspx?articleId=55426>
5. HealthEdge. (2026, February 27). Healthcare AI trends 2026: Payer vs. member readiness gap. HealthEdge.com. <https://healthedge.com/resources/blog/ai-in-2026-are-health-plans-and-members-aligned-on-the-future>
6. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., and Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, 33, 9459-9474.
7. Nixon Law Group. (2026, April 24). The 2026 guide to healthcare generative AI regulations: Frameworks and compliance for leaders. NixonLawGroup.com. <https://www.nixonlawgroup.com/resources/the-2026-guide-to-healthcare-generative-ai-regulations-frameworks-and-compliance-for-leaders>
8. Nogueira, R., and Cho, K. (2019). Passage re-ranking with BERT. *arXiv preprint arXiv:1901.04085*.
9. Prosper AI. (2026, May 20). AI automated claims management guide. GetProsper.ai. <https://www.getprosper.ai/blog/ai-automated-claims-management-definitive-guide>
10. Robertson, S., and Zaragoza, H. (2009). The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4), 333-388.